

Week 7 Draft Benchmark

Complex Models vs. the Week 6 Baseline

Mercer General Hospital · CariSurg MedTech Pathways · Interim Submission

Status: Interim draft. Full six-axis benchmark, interpretability write-up, cost-benefit memo, and decision journal are the Final Submission (Tuesday) — not included here.

Split: Same as Week 6 — 80/20 stratified, random_state=42 (44,096 train / 11,025 test). Reused exactly so the comparison is honest.

1. What was built this week

- Random Forest (n_estimators=300, class_weight="balanced") — untuned first pass.
- Random Forest, tuned via RandomizedSearchCV over n_estimators, max_depth, min_samples_leaf, max_features.
- Gradient Boosting (HistGradientBoostingClassifier, class_weight="balanced").
- Small MLP (two hidden layers, 64 then 32 neurons, scaled features).

Feature engineering added: shock_index (HR/SBP), pulse_pressure (SBP-DBP), spo2_rr_ratio, plus red-flag indicators for tachypnea, hypoxia, and fever. The tuned Random Forest's top features were triage_vital_sbp, shock_index, and triage_vital_dbp — engineered features earned a real place alongside the raw vitals.

2. Draft benchmark table — macro F1

Model	Macro F1
Small MLP	0.498
Baseline (Logistic Regression, Week 6)	0.492
Random Forest (tuned)	0.436
Gradient Boosting	0.416
Random Forest (untuned)	0.390

On macro F1 alone, none of the complex models clearly beat the Week 6 baseline — the Small MLP edges it out by 0.006, well within noise. This matters: complexity has to earn its cost, and macro F1 alone isn't showing that yet.

3. The number that actually matters: ESI-1 recall

Macro F1 treats every class equally, but the earlier Week 6 report showed accuracy-style metrics can hide how a model treats the rarest, most critical patients. Checking accuracy and ESI-1 recall on the same test set tells a very different story:

Model	Accuracy	Macro F1	Recall (ESI 1)
Baseline (LogReg)	0.671	0.492	0.188
Random Forest (tuned)	0.573	0.436	0.500
Gradient Boosting	0.550	0.416	0.312
Small MLP	0.638	0.498	0.312

The tuned Random Forest catches half of the truly critical patients (0.500) — nearly triple the baseline's 0.188 — despite losing ground on accuracy and macro F1. No model wins on every axis. This is precisely the accuracy-vs-interpretability-vs-recall trade-off the Week 7 materials describe, and it's the question the final cost-benefit memo has to answer with a recommendation, not just a table.

4. Compute notes (preliminary)

Full training-time and inference-time-per-prediction figures are a Final Submission task. For the interim, approximate fit times observed this week:

Model	Fit / search time
Gradient Boosting	~8s
Random Forest (untuned, single fit)	~40s
Random Forest (tuned, hyperparameter search)	~78s (12 fits)
Small MLP	~103s

These were measured on a compute-constrained single-core environment, so absolute numbers will differ on normal hardware — but the relative ordering (boosting fastest, MLP slowest) is expected to hold generally, and matches what the Tutorial 1 slides predicted about each model's cost profile.

5. Honest gaps for the final submission

- Random Forest's hyperparameter search was deliberately narrowed for this environment's compute (`n_iter=6`, `cv=2` instead of a fuller search) — worth re-running with a wider search before finalising a recommendation.
- No interpretability write-up yet — feature importances were printed, but "can I explain one prediction to Dr. Reyes in a minute" hasn't been answered for any model.
- No formal inference-time-per-prediction measurement (only training/fit time so far).
- No verdict yet. This table raises the trade-off; it doesn't resolve it — that's the cost-benefit memo's job.